CHAPTER 3

STRUCTURAL ORGANISATION IN ANIMALS

CHAPTER ESSENCE AT A GLANCE:
Histological specialization of animal tissues and the comparative organ systems of the Indian frog.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Epithelial Tissue Specialization:** Epithelia form protective coverings and secretory surfaces: simple squamous (diffusion boundary in lung alveoli and blood capillaries), simple cuboidal (absorption and secretion in proximal convoluted tubules with microvilli), simple columnar (stomach and intestinal lining), ciliated epithelium (directional mucus movement in bronchioles and fallopian tubes), and compound stratified epithelium providing mechanical protection against abrasion.

• **Cell Junction Architecture:** Epithelial integrity is maintained by specialized cell junctions: Tight junctions prevent solute leakage across epithelial sheets; Adhering junctions (desmosomes) cement neighbouring cells via cytoskeletal filaments; Gap junctions facilitate direct intercellular ionic and metabolic communication through cytoplasmic channels.

• **Connective and Muscular Tissues:** Connective tissues comprise loose areolar (subcutaneous support) and adipose (fat storage); dense regular (tendons connecting muscle to bone, ligaments connecting bone to bone); and specialized bone (osteocytes in lacunae) and cartilage (chondrocytes). Muscle tissues include striated voluntary skeletal fibers, unstriated involuntary smooth muscle, and branched intercalated cardiac muscle.

• **Frog Anatomy and Conduction Pathways:** The Indian frog (Rana tigrina) exhibits poikilothermy and cutaneous, buccopharyngeal, and pulmonary respiration. The 3-chambered heart (two atria, one ventricle) pumps mixed blood via conus arteriosus. In male frogs, 10 to 12 vasa efferentia emerge from testes, traverse kidneys, and enter Bidder's canal, joining the urinogenital duct opening into the cloaca.

ESSENTIAL PHYSIOLOGICAL PATHWAYS & STRUCTURES

$$\text{Cell Junction Functions} = \frac{\text{Tight (Leakage Barrier)} + \text{Adhering (Mechanical Cement)}}{\text{Gap (Rapid Intercellular Ionic Communication)}}$$

Functional Triad of Vertebrate Cell Junctions

$$\text{Frog Sperm Pathway} = \text{Testes} \rightarrow \text{10-12 Vasa Efferentia} \rightarrow \text{Kidney} \rightarrow \text{Bidder's Canal} \rightarrow \text{Cloaca}$$

Urinogenital Sperm Conduction in Male Rana tigrina

$$\text{Dense Connective Tissue} = \frac{\text{Tendon: Dense Regular (Connects Muscle to Bone)}}{\text{Ligament: Dense Regular (Connects Bone to Bone)}}$$

Tendon vs Ligament Anatomical Attachments

$$\text{Muscle Fibers} = \text{Skeletal (Striated and Voluntary)} + \text{Smooth (Spindle and Involuntary)} + \text{Cardiac (Branched with Discs)}$$

Structural and Functional Classification of Muscle Tissues

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** STRICT RATIONALIZED SYLLABUS GUARD: Invertebrate dissections were completely removed; only the morphology and organ systems of the Indian frog (Rana tigrina) are syllabus-active.

• **Alert:** In male frogs, the urinogenital duct conducts both sperm and urine, passing through Bidder's canal into the cloaca; in female frogs, oviducts and ureters open separately into the cloaca.

• **Alert:** Tendons connect skeletal muscles to bones (dense regular), whereas ligaments connect bones to bones.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Distinguish between tendons and ligaments based on the tissue types they connect. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• Tendons: Dense regular connective tissue cords that attach skeletal muscles to bones. **[1 Mark]**

• Ligaments: Dense regular connective tissue bands that attach one bone to another bone across joints. **[1 Mark]**

2. Describe the physiological functions of tight junctions, adhering junctions, and gap junctions. **[Score: 2 Marks]**

Answer & Key Valuation Steps:

• Tight junctions: Help to stop substances from leaking across an epithelial tissue sheet. **[0.75 Mark]**

• Adhering junctions: Perform cementing action to keep neighbouring cells together mechanically. **[0.75 Mark]**

• Gap junctions: Facilitate intercellular communication by connecting the cytoplasm of adjoining cells for rapid transfer of ions, small molecules, and second messengers. **[0.5 Mark]**

3. Differentiate between skeletal muscle, smooth muscle, and cardiac muscle based on striations, cell shape, branching, and voluntary control. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Skeletal muscle: Cylindrical, unbranched, multinucleated, striated, and under voluntary control. **[1 Mark]**

• Smooth muscle: Spindle-shaped (fusiform), unbranched, uninucleated, non-striated, and under involuntary control. **[1 Mark]**

• Cardiac muscle: Cylindrical, branched, uninucleated, striated with intercalated discs, and under involuntary control. **[1 Mark]**

4. Trace the step-by-step pathway of sperm conduction in a male frog (Rana tigrina) from the testes to the exterior, specifying the role of Bidder's canal. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Testes: Spermatozoa are produced in seminiferous tubules of paired testes. **[0.75 Mark]**

• Vasa efferentia: 10 to 12 vasa efferentia emerge from each testis and enter the adjacent kidney on its medial border. **[0.75 Mark]**

• Bidder's canal: Inside the kidney, vasa efferentia open into Bidder's canal, which connects to the longitudinal urinogenital duct. **[0.75 Mark]**

• Cloaca and exterior: The urinogenital duct carries both urine and sperm into the cloaca, exiting through the cloacal aperture. **[0.75 Mark]**

5. Explain how the Indian frog accomplishes cutaneous respiration in water and pulmonary respiration on land. **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Cutaneous respiration in water: Skin serves as an aquatic respiratory organ; dissolved oxygen in water diffuses across the thin, vascularized, moist skin directly into blood capillaries. **[1.5 Marks]**

• Pulmonary respiration on land: On land, the pair of elongated pink sac-like lungs acts as respiratory organs; air enters the buccal cavity through external nares and is pumped into lungs by floor-of-mouth movements. **[1.5 Marks]**

6. Describe the structural organization of compound epithelium. Why does it possess a limited role in secretion and absorption compared to simple epithelium? **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• Compound epithelium consists of two or more cell layers (multi-layered), with deeper dividing basal layers and flattened surface layers. **[1 Mark]**

• Its primary function is to provide protection against chemical and mechanical stresses (e.g., dry surface of the skin, moist lining of buccal cavity and pharynx). **[1 Mark]**

• Because it is multi-layered and thick, substances cannot easily diffuse across it, rendering its role in secretion and absorption limited compared to single-layered simple epithelium. **[1 Mark]**

7. A biopsy specimen reveals an epithelial layer with hair-like surface projections lining the inner surface of hollow organs, while another specimen from the intestinal lining reveals slender cells with basal nuclei. Identify each epithelial type, state their functions, and explain the clinical consequence if the ciliated epithelium is damaged. **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• First specimen identification: Ciliated columnar or cuboidal epithelium. Function: Moves particles or mucus in a specific direction over the epithelial surface (e.g., fallopian tubes, bronchioles). **[1.5 Marks]**

• Second specimen identification: Simple columnar epithelium. Function: Secretion of digestive enzymes and mucus, and absorption of digested nutrients. **[1.5 Marks]**

• Clinical consequence of ciliary damage: In respiratory bronchioles, damaged cilia cause accumulation of mucus, leading to chronic infections; in fallopian tubes, loss of ciliary movement prevents ovum transport, causing infertility or ectopic tubal pregnancy. **[1 Mark]**